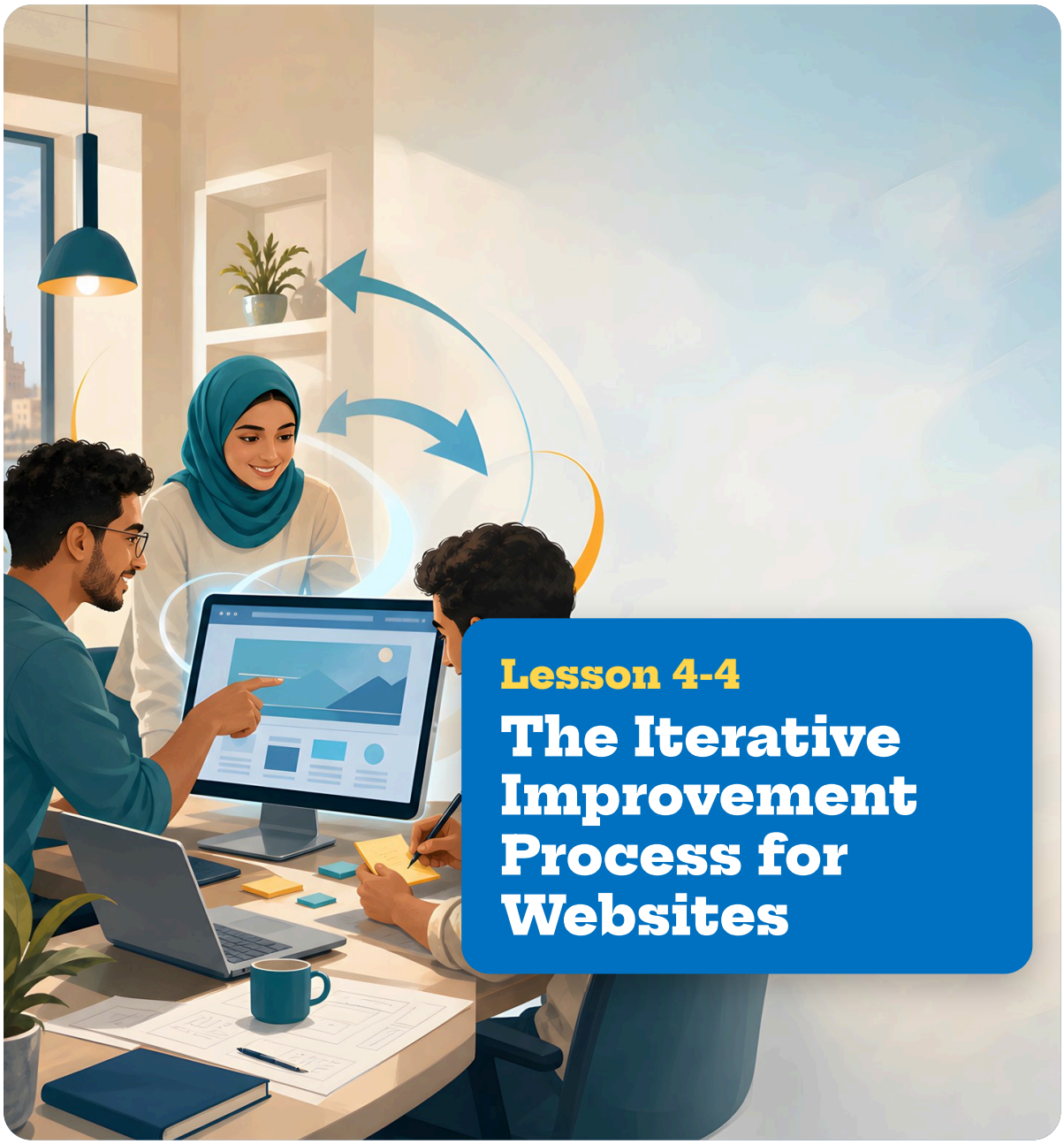


Lesson 4-4

The Iterative Improvement Process for Websites

Learning Objectives

- Explain the PDCA cycle and the steps of design thinking as ways of improving a website.
- Describe how A/B testing compares two options using real data, and explain why improvement is repeated in cycles.

A website is never truly finished. The first version is only a best guess, and a team learns what truly works only after real people use it. That is why good teams improve in loops: they plan a change, make it, measure the result, and act on what they learn — the PDCA cycle. They also design from the user's side (design thinking) and test real choices with data (A/B testing). Small, repeated improvements are better than waiting for one perfect launch.

? Guiding question: Why is improving a website a repeated loop rather than a one-time job, and how do teams decide what to change?

Point!

1. The PDCA Cycle

(1) **(PDCA cycle)**…a method that continuously advances improvement by repeating the following four steps. It is used widely in fields such as business improvement and quality management, not just websites.

Step	Description	Example in website improvement
Plan	Set goals and create a plan for improvement	Set the goal “reduce bounce rate by 10%” and plan a redesign of the homepage
Do	Execute according to the plan	Change the design of the homepage
Check	Measure results and evaluate the level of goal achievement	Measure the bounce rate after the change and evaluate whether the goal (10% reduction) was achieved
Act	Based on the evaluation, consider the next improvement measures	If the effect is sufficient, continue; if not, consider next steps such as revising headings or image placement

Key Fact:
Websites improve through repeated cycles: PDCA (Plan, Do, Check, Act) drives continuous improvement; design thinking finds problems from the user's side; A/B testing compares two options with real data.

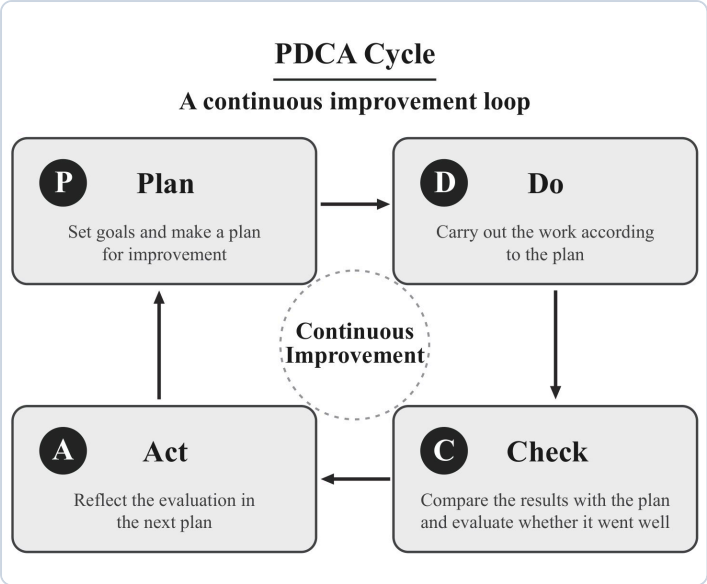
Key Concepts:

- PDCA cycle
- Plan / Do / Check / Act
- design thinking
- A/B testing

KEY TERMS
PDCA cycle — Plan→Do→Check→Act, repeated.
continuous improvement — getting better a little at a time.



Each Plan-Do-Check-Act cycle improves the site.



The PDCA cycle as a continuous improvement loop

(2) The important point of the PDCA cycle is not to aim for completion in one round, but to gradually improve through repetition.

2. Design Thinking

(1) **(Design thinking)**…a way of thinking that leads to discovering problems and generating solutions from the perspective of the user. It proceeds through the following five steps.

Step	Description
① **Empathize**	Put yourself in the user's position and understand their challenges and needs
② **Define**	Based on the information gleaned from empathy, clarify the problem to be solved
③ **Ideate**	Generate many ideas to solve the problem
④ **Prototype**	Quickly create a prototype based on the ideas
⑤ **Test**	Have users try the prototype and obtain feedback from them

(2) Based on test results, one may go back to the empathize or define stage. Improvement proceeds back and forth, not one-way.

PAUSE & THINK

Which PDCA step tells you whether your change actually worked?



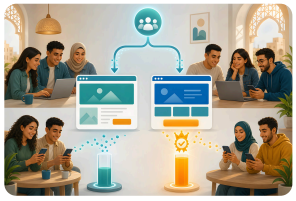
Design thinking: ideate, prototype, test, repeat.

3. A/B Testing

(1) **(A/B testing)**…a method in which two patterns (option A and option B) are prepared, each is shown to actual users, and data is examined to determine which is more effective.

e.g., Compare option A with a red purchase button and option B with a blue purchase button to see which gets clicked more.

(2) Key points of A/B testing



A/B testing compares two versions with real users.

① Change only one element at a time (if multiple elements are changed, it is unclear which change produced the effect).

② Make decisions only after collecting a sufficient amount of data.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A The PDCA cycle is a method that repeats the four steps Plan→Do→Check→Act.

B Design thinking is a way of thinking that solves problems from the developer's perspective.

C In A/B testing, it is important to change only one element at a time.

D The PDCA cycle is a method that is complete after one execution.

(2) For each of a–e, indicate which of the five steps of design thinking (A–E) it corresponds to.

a Understand the user's challenges and needs.

b Clarify the problem to be solved.

c Generate many ideas.

d Quickly create a prototype.

e Have users try the prototype.

A Empathize B Define C Ideate D Prototype E Test

✓ Solution

(1)

A Correct definition of the PDCA cycle. Therefore, ○.

B Design thinking is a way of thinking from the user's perspective. Therefore, ×.

C It is important to limit the changed element to one in A/B testing. Therefore, ○.

D The PDCA cycle is meaningful when repeated. Therefore, ×.

(2)

a: Understanding the user is empathize. **A**

b: Clarifying the problem is define. **B**

c: Generating ideas is ideate. **C**

d: Creating a prototype is prototype. **D**

e: Having users try is test. **E**

KEY TERMS

design thinking — solve from the user's view in five steps.

A/B testing — compare two options with real data.

💡 PAUSE & THINK

In A/B testing, why change only one element at a time?

Try

1 Answer the following questions.

1. What is the term for the improvement method that repeats the four steps Plan→Do→Check→Act?
2. What is the term for the way of thinking for discovering problems and generating solutions from the perspective of the user?
3. What is the term for the method in which two patterns are prepared and compared with data to see which is more effective?
4. Name the five steps of design thinking in order.

2 Answer the following questions.

(1) From the following options (A–D), choose the one that corresponds to "Check" in the PDCA cycle.

A Set the goal of reducing the bounce rate by 10%.

B Change the design of the homepage.

C Measure the bounce rate after the change.

D Consider another improvement measure.

(2) From the following options (A–D), choose the one that is NOT an appropriate way of conducting A/B testing.

A Prepare two patterns and compare them.

B Change multiple elements at the same time.

C Make decisions only after collecting a sufficient amount of data.

D Show each pattern to actual users.

Think as an Engineer — Investigate & Decide

Your school club runs a web page for its yearly event, but few students sign up through it. You lead one round of improvement using the PDCA cycle.

☐ **Redesign everything at once** ☐ **Improve in one measured PDCA loop**

1 · Collect data (Check first). Decide one number you could measure now — for example, how many of ten classmates find the "sign up" link within five seconds. Record the result as your starting point.

2 · Plan one change. From the users you watched, name the single biggest obstacle (for a specific group, such as a student opening the page on a small phone screen) and plan one change to fix it. State the goal as a number (e.g., "8 of 10 find it in five seconds").

EXAM-STYLE QUESTION

Explain why a website team improves in repeated PDCA cycles instead of trying to build a perfect site in one go. **[6]**
Refer to: measuring results and acting on them.

PAUSE & THINK

Match each option above to a PDCA step: which is Plan, Do, Check, Act?

3 · Do, Check, Act — and decide. Describe how you would make the change, measure again, and decide the next step. Choose one element to settle with an A/B test, and say why you change only that one element. **In pairs,** compare plans and agree on the single change to try first.

Give evidence for your answer.

Stuck? Start from your measured number (Check), pick the one obstacle it exposes, plan a single fix (Plan), then loop — do not change everything at once.

🌍 In a New Context

An online shop finds that its “buy” button is rarely clicked. Describe how you would use one A/B test (changing only one element) and one PDCA loop to improve it, and say which number would tell you it worked.

❓ Lesson Question — Answered

The lesson asked why improving a website is a repeated loop and how teams decide what to change. A first version is only a guess, so teams improve continuously rather than once. The PDCA cycle repeats four steps — Plan (set a goal and plan a change), Do (make it), Check (measure the result), and Act (decide the next step) — and its point is not completion in one round but gradual improvement through repetition. To decide what to change, design thinking works from the user's side through five steps (Empathize, Define, Ideate, Prototype, Test), moving back and forth rather than one-way, while A/B testing compares two options with real data, changing only one element at a time and deciding only after enough data. Together they make improvement small, measured, and repeatable.

KEY TERMS

prototype — a quick, testable draft.

iterate — repeat the loop to improve.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

As a method for improving websites, there is the (a) cycle. It repeats four steps: (b), in which goals are set; (c), in which the plan is executed; (d), in which results are measured; and (e), in which improvement measures are considered. Also, (f), in which two patterns are prepared and compared with data to see which is more effective, is a useful method.

1. Fill in the blanks (a)–(f).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes design thinking.

A A method that uses numerical data to analyze website usage.

B A way of thinking that leads to discovering problems and generating solutions from the perspective of the user.

C A method in which experts identify UI problems based on rules of thumb.

D A method of creating a blueprint that roughly shows the layout of a web page.

(2) From the following options (A–D), choose the one that most appropriately describes the concept of the website improvement process.

A If a perfect website is built initially, no improvement is needed.

B It is enough to improve only the visual design.

C It is important to gradually improve by repeating evaluation and improvement.

D All improvements should be done with A/B testing alone.

⭐ REFLECT & CHALLENGE

Reflect: Think of a website you use often. Which one small change do you think would help most, and how would you measure whether it worked?
Challenge: You may run only one A/B test this month on the event page. Choose the single element to test, predict which option will win and why, and say what number would prove you right or wrong.

⭐ Key takeaway

Remember:
Improve in loops: PDCA (Plan, Do, Check, Act) for continuous improvement, design thinking (Empathize, Define, Ideate, Prototype, Test) from the user's side, and A/B testing to settle one change with data.